

Tutorial on speechbrain

Why Speech brain?

- What is speechbrain?
 - Why need of such a tool ?
 - What advantages it offer ?
-
- Speech brain is a tool built over many SOTA approaches for speech processing .
 - Uses hugging face models trained over large amount of datasets , and users can fine tune on their datasets.

Use cases

- **Speech recognition**
- **Source separation**
- **Speaker recognition**
- **Speech enhancement**
- **Speech processing**
- **Speech enhancement**

There are lot more. All these models are inbuilt in speech brain and can be directly imported and used.

Installation

- <https://speechbrain.readthedocs.io/en/latest/installation.html#:~:text=Install%20locally,-Once%20you%20have&text=git%20clone%20https%3A%2F%2Fgithub.txt%20pip%20install%20%2D%2Deditable%20.&text=Any%20modification%20made%20to%20the.with%20the%20%2D%2Deditable%20flag>

Both methods work fine - pip installation and using google colab or downloading it locally .

Main components of speechbrain

Brain class

1. This class is nothing but the training of any general neural network. In deep learning key component is to train the model by defining an objective function and the updating the weights to minimise the loss. Same thing is done by brain class.
 2. Whenever we make a brain class we have to define out two functions -
 - a. `compute_forward` -> this function defines the model that you want to use (eg linear layers , recurrent layers etc.)
 - b. `compute_objectives` -> loss function that you want to use (eg L1 loss , MSE loss etc.)
- Rest all the things such as calculating gradients , updating weights are all handled in original implementation.
 - we sub-class the class(SimpleBrain) and override the methods . Finally we do `fit()` to fine tune our model according to parameters we defined.
 - https://colab.research.google.com/drive/1fdqTk4CTXNcrcSVFvaOKzRfLmj4fJfwa?usp=sharing#scrollTo=F_CemYaLIB

HyperPyYAML

- This is a .yaml file for storing hyperparameters and other metadata that can be used by our model .
- <https://colab.research.google.com/drive/1Pg9by4b6-8QD2iC0U7lc3Vxq4GEwEdDz?usp=sharing>

Data Loading Pipeline

Parsing

- First step is to parse the dataset we have , in json or csv format. Parsing will create objects(with unique id) with key value pair in it.

Dynamic Item Dataset

- So after parsing we have to create our own dataset such that items that the user wants to be returned must be specified in some way that user .
- Let's say we have keys as ['file_path', 'words', 'spkID', 'length'] , we want to add audio signal corresponding to file path in this object , so we can add a dynamic item audio signal in this object.

Dynamic Item Pipelines (DIPs)

- This task is handled by specifying **Dynamic Item Pipelines** for each dynamic item the user wants to get from the dataset.
- The user can specify an arbitrary number of pipelines.

CategoricalEncoder

- This part is used to encode the strings to integers. For example let's say in our json dataset we have speaker_id: spk1 . So we actually want to have a number assigned to spk1 , spk2 .. . So categorical encoder encodes word2ind and ind2word .

Complete data flow pipeline

<https://colab.research.google.com/drive/1AiVJZhZKwEI4nFGANKXEe-ffZFvXKwH?usp=sharing#scrollTo=YoLjI4HtGlgC>

- Now we will see two use cases of speechbrain -

- 1. How to fine tune the inbuilt models like asr , source separation etc . on our own dataset .**
- 2. How to extract speech features using speechbrain.**

These are two main things we will require while using speech brain in doing projects . Obviously more and more can be explored.

Fine tuning pretrained models -

- <https://colab.research.google.com/drive/1LN7R3U3xneDgDRK2gC5MzGkLysCWxuC3?usp=sharing#scrollTo=EfpGJ3FlgFRu>

Feature extraction using speechbrain

- Speech brain can be used to extract a large number of important features in speech signal that can be analysed further. Some of them are -
 - Short Term Fourier Transform(STFT) -> used to determine the sinusoidal frequency and phase content of local sections of a signal as it changes over time.
 - Spectral magnitude -> magnitude of each spectrum
 - Filter banks -> Filters are applied over spectrogram . 40-80 filters are applied with more on lower side to capture more relative information.compresses the rich information present .

- Mfcc -> further compress the information . Discrete cosine transform applied over filter banks.
- Context window
 - Derivatives - directly subtracting the consecutive mfcc values. First and second order derivatives are called delta and delta-delta respectively. Total 39 dimension feature vector (13 static , 13 delta , 13 delta-delta)
 - Context window - improvement of context by concatenating some window size number of consecutive features in a 39 dimensional feature vector to learn more context.

- https://colab.research.google.com/drive/1CI72Xyay80mmmagfLalleRoDgswWHT_g?usp=sharing
- <https://github.com/speechbrain/speechbrain/blob/develop/speechbrain/processing/features.py>